# Renewable and Nonrenewable Resources

Chapter 1

Three hikers prepare dinner at a campsite. One boils pasta over the blue flame of a propane gas stove. Another fries fish over a wood fire. A third roasts vegetables in a solar oven over the course of the afternoon.

The wood and solar energy used for cooking are **renewable resources**. They will not be used up completely. While neither resource is available everywhere or all the time, both are renewable. The sun comes up every day. Trees can continually be planted and harvested.

The propane in a camping stove is a **nonrenewable resource**. It takes much longer than a human lifetime for propane and other fossil fuels to form.

## Big Question

What is the difference between renewable and nonrenewable resources?

## Vocabulary

**renewable resource, n.** a resource that can be restored more quickly than it is used up

**nonrenewable resource, n.** a resource that cannot be restored as quickly as it is used



**Propane is abundant now. But once the natural gas that is used to make propane is all used up, more will not form fast enough for it to be used as an energy resource.**

## Availability of Energy Resources Varies

Natural gas is abundant. So are petroleum and coal, two other types of **fossil fuels**. Abundant does not mean renewable, however. What makes a resource renewable or nonrenewable is how quickly it is used compared to how long it takes to become available.

**Vocabulary**

**fossil fuel, n.** a fuel formed from the fossilized remains of organisms

Fossil fuels develop over thousands or millions of years. Humans have been extracting fossil fuels from Earth's crust for centuries. But the rate at which we use them is much faster than the rate at which they form. That is why they are nonrenewable even though they are plentiful right now.

A resource can also be renewable but not necessarily abundant. For example, some locations do not get enough wind to generate electricity reliably for people's needs. Even sunlight can be too scarce to serve as a dependable energy source in some locations.



**In a sunny location, sunlight is both renewable and abundant. In Antarctica during the winter, the sun only dimly lights the sky for a few hours a day.**

## Some Energy Resources Are Renewable

Renewable resources that are used for energy include wood, wind, flowing water, sunlight, and heat from Earth's interior. In some cases, these energy resources are used in their original forms to apply energy or force to something. For example, sails on a sailboat harness wind to push the boat across the water. In other cases, the energy is transformed to a new form. A windmill, for example, converts the energy of moving air to mechanical energy that turns gears in machinery. These days, most energy resources are converted to **electricity**.

### Vocabulary

**electricity, n.** a form of energy resulting from the flow of charged particles









**Old Mills:** The windmill and water mill were both used to convert wind energy into mechanical energy to grind grain into flour.

**New Turbines:** This wind turbine and this water turbine convert wind and water energy into mechanical energy. A generator converts the mechanical energy into electricity.

## Renewable Resources Can Still Be Costly

The use of renewables is not necessarily problem free. For example, large wind turbines built to harness wind power can pose hazards to birds as well as aircraft. Renewable energy technology can also be very expensive. A homeowner who wants to turn his or her roof into a solar power generator may have to spend thousands of dollars to install solar panels.

Tapping renewable energy resources can also be harmful to landscapes, rivers, and other habitats. The dams that provide hydroelectric power in some locations may damage the environment around them. A dam may block fish such as salmon from traveling upstream to reproduce. The construction of a dam usually involves flooding a large area of land.



**The Three Gorges Dam produces an enormous amount of electrical energy. But it also displaced more than a million people.**

## Petroleum Is a Nonrenewable Resource

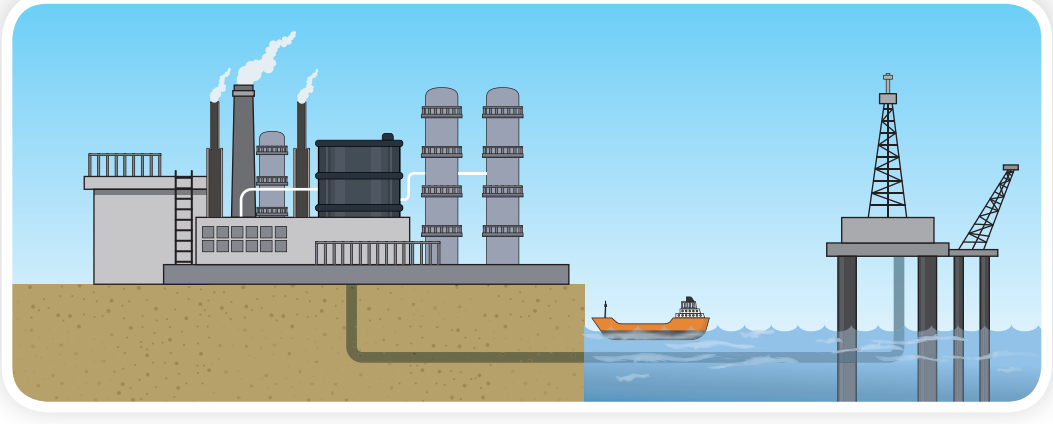
**Crude oil**, also known as petroleum, is a liquid formed over time by fossil remains that are under intense pressure. Petroleum is refined into many liquid fuels, including gasoline, diesel, and kerosene. Petroleum is also made into plastics, lip balm, and other products.

**Vocabulary**

**crude oil, n.** a liquid also known as *petroleum*

Petroleum is taken from Earth's crust by drilling holes deep into the rock. A large underground field of petroleum may be under enough pressure to force the petroleum out of the crust and up through a pipe. Petroleum can also be pumped from the ground. The hole through which the petroleum is taken is called a well. If a well is not controlled, it can blow out, causing an oil spill. Oil spills are dangerous to humans working on or near the oil well. They also can harm wildlife and ecosystems. These are some of the costs of using petroleum.

Petroleum can be transported by pipelines, trains, trucks, ships, and other forms of transportation. It is processed at a refinery, a place where crude oil can be changed into many different useful products. These are the benefits of using petroleum.



**Much petroleum is pumped out of the seafloor. Oil spills can occur when oil wells blow out, pipelines rupture, or oil tankers have accidents at sea.**

## Natural Gas and Coal Are Nonrenewable Resources

Natural gas is another fossil fuel. It often forms in the same type of location as petroleum, from similar remains of organisms. Like crude oil, gas can be extracted from Earth's crust and transported to other places. And like crude oil (petroleum), it must be refined before it can be used. Natural gas is used in many homes for cooking and heating. This requires a network of pipes to bring the gas to homes. Because gas is under pressure and flammable, accidents such as explosions can occur.

Coal is the third major type of fossil fuel. Unlike gas and petroleum, coal is a solid. It forms mainly from the fossil remains of plants that once lived in very wet habitats. Coal has been abundant and inexpensive for centuries. But mining it is dangerous. When coal is burned, it produces a lot of carbon dioxide and releases other pollutants into the atmosphere.

These fossil fuels are usually burned to transform chemical energy into thermal energy, heat. The transformation can cause the parts of a car engine to move or heat water inside an electric generator.

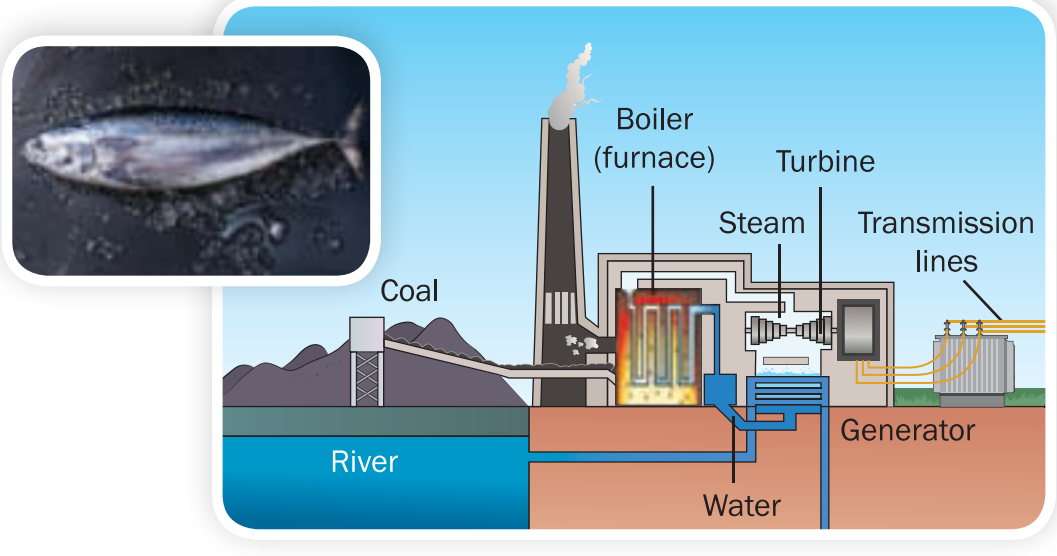


**A piece of coal takes millions of years to form but burns up in minutes.**

## Using Coal Has Costs and Benefits

Many power plants that supply electricity are powered by coal. In them, coal is burned below a large container of water called a boiler. Burning coal heats the water, which boils and forms steam. Because the boiler is sealed, the steam builds up. As the steam expands, it moves through pipes to an area of lower pressure. Here, the moving steam spins turbines. These turbines convert the energy of the steam into mechanical energy that moves a generator. This converts the energy of motion into electricity.

The sooty gas produced by the burning coal can be treated to reduce some of the pollution. In some locations, the carbon dioxide is directed underground. In general, however, most of the carbon pollution goes into the atmosphere. Air pollution from coal-fired power plants can also contain mercury, which can end up in the water. It accumulates in organisms that humans eat, such as tuna. These are some of the costs of using coal.

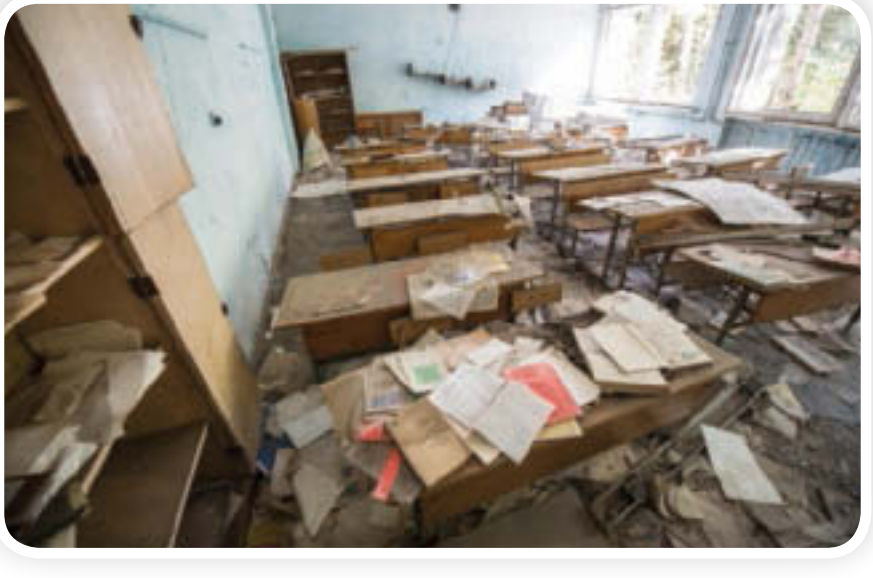



**Burning coal produces carbon dioxide and mercury pollution. Mercury can accumulate in marine organisms such as fish and then end up in humans.**

## Nuclear Power Is a Nonrenewable Resource

Another type of nonrenewable energy people use to produce electricity is nuclear energy. Nuclear energy has been used to make weapons, but it has also been used in nuclear power plants. In a nuclear power plant, the release of energy is controlled. Controlling the energy release allows heat to be directed into a boiler similar to what is used in a coal-powered plant. The boiler produces steam that spins a turbine, which makes a generator produce electricity.

Nuclear power does not release carbon dioxide into the environment, but it does have costs. If the release of energy is not controlled, the reactor can generate too much heat and melt part of the reactor. This exposes the surrounding area to radiation. Radioactivity is the release of harmful particles from a substance. Radioactivity can be deadly to organisms. It can also linger in an environment and cause a variety of diseases over a very long time.



**In 1986, an accident occurred at a nuclear power plant near the Ukrainian city of Chernobyl. People for many miles around had to abandon the area to escape exposure to radiation. The area will not become safe again within the lifetimes of the people who had to leave.**